PART 3 · MODULE D

Advanced development: back end, security, performance, testing and deployment

The advanced topics that turn a demo into a product a business can depend on.

These topics apply across the ten professions. Read each section, then use one of them in a project from the chapters above.

D1. Backend development and APIs

The back end receives requests, checks them, applies business rules, reads or writes the database and returns a response. Keep endpoints small and named after the resource (`/api/bookings`). Return the right status code (see Part 1, Module A).

D2. Authentication and authorisation

Authentication proves who the user is; **authorisation** decides what they may do. Use a tested service (for example Supabase Auth, Firebase Auth or a framework's built-in library) rather than writing your own. Store passwords only as hashes. Check permissions on the server for every request.

D3. Database management

Design tables around the questions the app asks. Add indexes to columns you search often. Back up regularly and test a restore. Never give the public app the database administrator key.

D4. API integration

Read the provider's documentation, store keys in environment variables, handle timeouts and errors, and respect rate limits. Use webhooks for events that happen later (for example a payment confirmed).

D5. Payment gateways

Use the provider's hosted checkout or tested SDK; never store card numbers yourself. Confirm payment with a server-side webhook, not only the browser redirect. Test in the provider's test mode first (see the E-commerce chapter).

D6. Website security

- Use HTTPS everywhere and keep software updated.
- Validate input on the server; escape output to prevent script injection.
- Set secure cookies and a content security policy where the platform allows.
- Keep secrets out of code and out of public repositories.

D7. Performance optimisation

Measure first (Lighthouse, browser performance panel). Compress and size images, load scripts only where needed, cache static files, and avoid unnecessary database queries (for example, load a list once instead of once per row).

D8. Testing

Test the rules that matter most: money calculations, permissions, form validation. The Node.js built-in test runner, a browser test tool (Playwright) and manual checks on a phone cover most small projects.

D9. Deployment, cloud hosting and maintenance

Deploy from git, with a staging copy for testing. Use managed hosting for static sites and serverless functions (Vercel, Netlify, GitHub Pages), and a managed database. Monitor errors, renew domains and certificates, and update dependencies on a schedule. Write a one-page handover for the client.

Part of **Master Freelancing: Development and IT**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.